ENSIA

Artificial Intelligence and Data Science
Semester 7 - 2026/2027

Introduction to Mobile Robotics

Lab Assignments

Installation and Setup of ROS 2 Jazzy
Ubuntu 24.04 LTS (Noble Numbat)

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Prerequisites

Before proceeding, make sure that you have:

- Ubuntu 24.04 LTS (Noble Numbat) installed.
- An active Internet connection.
- A user account with administrative privileges (`sudo`).
- A terminal application.

Note: ROS 2 Jazzy Jalisco is the ROS 2 distribution recommended for this guide. It is designed for Ubuntu 24.04 LTS.

1 System Setup

Follow the official ROS 2 installation instructions:

https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html

1.1 Set the Locale

ROS 2 requires a UTF-8 locale. First, check the current locale configuration:

```
locale
```

Install the locale utilities and configure the English UTF-8 locale:

```
sudo apt update
sudo apt install locales

sudo locale-gen en_US en_US.UTF-8

sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8

export LANG=en_US.UTF-8
```

Verify the configuration:

```
locale
```

Make sure that the locale variables use UTF-8 encoding.

1.2 Enable the Required Repositories

First, install the tools needed to configure the Ubuntu repositories:

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

1.3 Add the ROS 2 Repository

Install `curl` and update the package index:

```
sudo apt update
sudo apt install curl -y
```

Download the ROS 2 repository signing key:

```
sudo curl -sSL \
  https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
  -o /usr/share/keyrings/ros-archive-keyring.gpg
```

Add the ROS 2 repository to the APT sources list:

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive
    -keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo
    $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

Update the package index:

```
sudo apt update
```

Important: On Ubuntu 24.04, the expected codename is `noble`. Verify it using:

```
. /etc/os-release
echo $UBUNTU_CODENAME
```

2 Install ROS 2 Jazzy

Update the package index and install available system updates:

```
sudo apt update
sudo apt upgrade
```

If prompted, review the proposed package changes before confirming the upgrade.

2.1 Install the Desktop Version

Install the ROS 2 Jazzy desktop distribution:

```
sudo apt install ros-jazzy-desktop
```

The desktop installation includes the ROS 2 core packages, RViz, demos, and other useful tools for robotics development.

2.2 Install Development Tools

Install the ROS development tools:

```
sudo apt install ros-dev-tools
```

These tools support the development, building, and maintenance of ROS 2 packages.

3 Configure the ROS 2 Environment

3.1 Source the Setup File

To use ROS 2 commands, source the environment setup file:

```
source /opt/ros/jazzy/setup.bash
```

This command must be executed in each new terminal unless the setup file is automatically sourced.

3.2 Automatically Source ROS 2

To avoid manually sourcing the setup file each time a new terminal is opened, add the command to the Bash startup file:

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

Open a new terminal or reload the configuration:

```
source ~/.bashrc
```

Note: Add this line only once to avoid duplicate entries in your `.bashrc` file.

3.3 Verify the Environment Variables

Check the ROS environment variables:

```
printenv | grep -i ROS
```

The output should contain variables similar to the following:

```
ROS_VERSION=2
ROS_PYTHON_VERSION=3
ROS_DISTRO=jazzy
```

The exact output may contain additional variables.
If `ROS_DISTRO` is missing or shows an unexpected distribution, source the correct setup file:

```
source /opt/ros/jazzy/setup.bash
```

If the problem persists, verify that ROS 2 Jazzy was installed correctly before proceeding.

4 Test the Installation

In this section, two ROS 2 nodes will be executed to verify that the installation works correctly. A C++ talker publishes messages, and a Python listener receives them.

4.1 Run the Talker Node

Open a terminal and execute:

```
source /opt/ros/jazzy/setup.bash
ros2 run demo_nodes_cpp talker
```

The terminal should display messages indicating that data is being published.

4.2 Run the Listener Node

Open a second terminal and execute:

```
source /opt/ros/jazzy/setup.bash
ros2 run demo_nodes_py listener
```

The listener should display messages received from the talker.

4.3 Expected Result

If the installation is successful:

- The talker publishes messages continuously.
- The listener receives and displays those messages.
- Both nodes execute without errors.

This test verifies basic ROS 2 communication using the C++ and Python APIs.

5 Uninstall ROS 2 Jazzy

If you need to remove ROS 2 Jazzy, execute:

```
sudo apt remove '~nros-jazzy-*'
sudo apt autoremove
```

Warning: Review the packages that APT proposes to remove before confirming the operation.

To remove the ROS 2 repository configuration and signing key:

```
sudo rm /etc/apt/sources.list.d/ros2.list

sudo rm /usr/share/keyrings/ros-archive-keyring.gpg

sudo apt update
```

If you added the ROS environment setup command to your `.bashrc` file, remove that line as well.
Removing ROS 2 does not automatically remove the rest of your Ubuntu installation or unrelated packages.

6 Additional Resources

For further information, consult the official documentation:

- ROS 2 Jazzy documentation: https://docs.ros.org/en/jazzy/
- Installing ROS 2 using Debian packages: https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html
- ROS 2 tutorials: https://docs.ros.org/en/jazzy/Tutorials.html

7 Installation Guidelines for Different Operating Systems

Important Instructions

Attached is a guide for installing ROS 2 Jazzy on Ubuntu 24.04 LTS. **Please ensure that you complete the installation and verify that ROS 2 is working correctly before the lab session.**

The recommended configuration for this course is:

- Operating system: Ubuntu 24.04 LTS (Noble Numbat).
- ROS distribution: ROS 2 Jazzy Jalisco.
- Installation type: ROS 2 Desktop.

If you are using Windows, macOS, or a Linux distribution other than Ubuntu 24.04, you can consider one of the alternative installation methods described below.
Important: Some of the resources listed below were originally written for ROS 2 Humble and Ubuntu 22.04. When possible, use ROS 2 Jazzy with Ubuntu 24.04 to match the main installation guide. If you choose an alternative configuration, verify that it is compatible with the software used in the lab assignments.

Official Documentation

For additional information, consult the official ROS 2 installation documentation:

- ROS 2 Jazzy installation on Ubuntu: https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html
- ROS 2 Humble installation on Ubuntu 22.04 (for reference or when Humble is specifically required): https://docs.ros.org/en/humble/Installation/Ubuntu-Install-Debs.html

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7.1 Option 1: Distrobox (Linux Users)

Distrobox is a container management tool that allows you to run a different Linux distribution inside your existing operating system. It provides integration between the container and the host system and uses Podman or Docker as its container backend.

Distrobox can be a lightweight alternative to a virtual machine for users who cannot install ROS 2 directly on their host operating system.

Installation instructions

1. Install Distrobox by following the official instructions:

   https://github.com/89luca89/distrobox

2. Create a container based on a compatible Ubuntu release, preferably Ubuntu 24.04 for ROS 2 Jazzy.
3. Install ROS 2 inside the container by following the official Ubuntu installation instructions provided in this document.
4. Verify the installation by running the ROS 2 talker and listener examples.

For additional assistance with Distrobox setup, you may find the following video helpful:
https://www.youtube.com/watch?v=eiDt4O6UPRw

Note: GUI applications, networking, hardware access, and simulator integration may require additional configuration inside a container.

7.2 Option 2: UTM (macOS Users)

UTM allows you to run virtual machines on macOS. It is particularly useful for students who need a Linux environment on a Mac.

Installation instructions

1. Download and install UTM from its official website:
   https://mac.getutm.app/
2. Download the Ubuntu 24.04 LTS Desktop image corresponding to your computer's architecture.
   - For Intel-based Macs and compatible 64-bit PC architectures, consult:
     https://releases.ubuntu.com/noble/
   - For Apple Silicon Macs (ARM64), consult:
     https://cdimage.ubuntu.com/releases/noble/release/
3. Create a virtual machine in UTM and install Ubuntu.
4. Once Ubuntu is installed, follow this guide to install ROS 2 Jazzy.

The following resources may also be helpful:

- Guide to installing ROS 2 on Ubuntu 22.04 on an M1 Mac:
  https://www.theroboticsspace.com/blog/How-To-Install-ROS-2-in-Ubuntu-22-04-On-M1-Mac/
- Video guide for setting up Ubuntu in a virtual machine:
  https://www.youtube.com/watch?v=6sLPp_szbs4

Note: The linked tutorial uses Ubuntu 22.04 and ROS 2 Humble. For this course, use Ubuntu 24.04 and ROS 2 Jazzy whenever possible. The Ubuntu image must match the virtual machine's CPU architecture.

7.3 Option 3: WSL 2 (Windows Users)

Windows Subsystem for Linux (WSL) allows you to run a Linux environment directly on Windows without installing a traditional virtual machine manually.

WSL 2 is a convenient option for students who want to use Linux-based ROS 2 tools while keeping Windows as their primary operating system.

Installation instructions

1. Install WSL by following Microsoft's official instructions:

   https://learn.microsoft.com/en-us/windows/wsl/install

2. Install Ubuntu 24.04 LTS under WSL 2.

3. Open the Ubuntu terminal and follow the ROS 2 Jazzy installation instructions in this document:

   https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html

4. Verify the installation using the talker and listener examples.

Important: WSL 2 may require additional configuration for graphical applications, USB devices, networking, and robot simulators. Make sure that the tools required for your lab assignments work correctly in your environment.

7.4 Option 4: RoboStack (Linux, macOS, and Windows)

RoboStack provides ROS and ROS 2 packages through the Conda ecosystem. It offers an alternative installation method for systems on which a native ROS 2 installation may be inconvenient.

RoboStack provides a ROS 2 Jazzy environment for several platforms, including Apple Silicon Macs.

Installation instructions

1. Consult the official RoboStack documentation:

   https://robostack.github.io/GettingStarted.html

2. Follow the installation instructions for your operating system.

3. Select the ROS 2 Jazzy environment, where supported by your platform.

4. Install the packages needed for your lab assignments and verify that the required ROS 2 commands are available.

For additional information about installing ROS 2 Humble and Gazebo on Apple Silicon Macs, consult:

https://medium.com/@shubhjain_007/installing-ros2-humble-gazebo-on-mac-m1-m2-m3-silicon-8

Note: The linked tutorial concerns ROS 2 Humble. For this course, prefer the RoboStack Jazzy environment when it provides the packages you need.

7.5 Option 5: Multipass (Linux, macOS, and Windows)

Multipass allows you to create and manage Ubuntu virtual machines on Linux, macOS, and Windows.

It is an alternative for students who prefer to work in an isolated Ubuntu environment without replacing their existing operating system.

Installation instructions

1. Install Multipass by following the official installation instructions:

   https://multipass.run/

2. Consult the following guide for setting up a ROS development environment:

   https://ubuntu.com/blog/ros-development-on-linux-windows-and-macos

3. Create an Ubuntu virtual machine and install ROS 2 Jazzy by following the official installation instructions in this document.

Important: If you use a Multipass blueprint that installs a specific ROS distribution, make sure that the blueprint supports ROS 2 Jazzy and Ubuntu 24.04.

Example: Launching a Virtual Machine Choose a unique name for your virtual machine. For example:

```
multipass launch --name jazzy-vm-your_name \
    24.04
```

Replace `your_name` with a suitable identifier.
After the virtual machine has been created, open a shell:

```
multipass shell jazzy-vm-your_name
```

Then install ROS 2 Jazzy by following the main installation guide.

Troubleshooting SSH Key Errors If you encounter a "No such file or directory" error when configuring an SSH key, follow these steps.

Step 1: Check for an existing SSH key.

```
ls -al ~/.ssh
```

Public key filenames typically end with `.pub`, for example, `id_rsa.pub` or `id_ed25519.pub`.

Step 2: Generate an SSH key if necessary.

If no suitable key exists, create one using:

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

Alternatively, you can generate an RSA key:

```
ssh-keygen -t rsa -b 4096 \
    -C "your_email@example.com"
```

When prompted for the file in which to save the key, press Enter to accept the default location.
Note: Generating an SSH key is only necessary if your chosen workflow requires one.

7.6 Option 6: VirtualBox (Linux, Windows, and macOS)

VirtualBox allows you to run Ubuntu in a virtual machine without changing the host operating system.

This option is suitable for students who prefer a conventional virtual machine setup.

Installation instructions

1. Download VirtualBox from its official website:

   https://www.virtualbox.org/

2. Download the Ubuntu 24.04 LTS Desktop image appropriate for your computer's architecture:

   https://releases.ubuntu.com/noble/

3. Create a virtual machine, install Ubuntu, and configure the necessary resources.

4. Follow the ROS 2 Jazzy installation instructions provided in this document.

For Windows users, the following tutorial explains how to install ROS 2 in an Ubuntu 22.04 virtual machine:
https://www.theroboticsspace.com/blog/How-To-Install-ROS-2-in-Ubuntu-22-04-VM-On-Windows/
Note: The linked tutorial uses Ubuntu 22.04 and ROS 2 Humble. Use Ubuntu 24.04 and ROS 2 Jazzy for the configuration recommended in this course.

Final Checklist Before the Lab Session

Before attending the lab session, make sure that you have completed the following tasks:

- ☐ Installed ROS 2 Jazzy or an alternative environment compatible with the lab requirements.
- ☐ Configured the ROS 2 environment.
- ☐ Verified that the `ROS_DISTRO` environment variable is set correctly.
- ☐ Successfully executed the ROS 2 talker and listener examples.
- ☐ Confirmed that the required terminal commands work correctly.
- ☐ Checked that the chosen environment supports the tools and simulators required for the upcoming lab assignment.

Please complete these preparations before the lab session. This will allow us to focus on the robotics exercises rather than troubleshooting installation problems during class.